ASSIGNMENT 1:

Mansha Bhatnagar

1024150192

2O23

(Q1) Write a program (WAP) to display "Hello World" on console display.

CODE:

```
Welcome        ass1.cpp 3 ×
C: > Users > HP > ass1.cpp > main()
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World!";
    return 0;
}
```

OUTPUT:

```
[Running] cd "c:\Users\HP\" && g++ ass1.cpp -o ass1 && "c:\Users\HP\"ass1
Hello World!
[Done] exited with code=0 in 1.537 seconds
```

(Q2) WAP to that reads two values from keyboard and displays the larger value on the screen.

CODE:

```
E: > ass1.cpp > main()
#include <iostream>
using namespace std;


int main() {
    int a, b;
    cout << "Enter first number: ";
    cin >> a;
    cout << "Enter second number: ";
    cin >> b;
    cout << "The larger number is: " << ((a > b) ? a : b);

    return 0;
}
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\HP> cd "e:\" ; if ($?) { g++ ass1.cpp -o ass1 } ; if ($?) { .\ass1 }
Enter first number: 4
Enter second number: 5
The larger number is: 5
PS E:\>
```

(Q3) WAP to read three values a, b, c from keyboard and display the value of x = a/b-c.

CODE:

```
E: > ass1.cpp > main()
#include <iostream>
using namespace std;

int main() {
    float a, b, c, x;

    cout << "Enter the numbers: ";
    cin >> a >>b >>c;

    x = a / (b - c);

    cout << "Value of x = " << x;

    return 0;

}
```

OUTPUT:

```
PS E:\> cd "e:\" ; if ($?) { g++ ass1.cpp -o ass1 } ; if ($?) { .\ass1 }
Enter the numbers: 3
9
6
Value of x = 1
```

(Q4) WAP to implement the following control characters:

-'\n' is for new line, or you can use endl – cout<<endl<<"message";

- '\t' is for tab - '\a' is an alarm sound

- '\r' is carriage return to go to the beginning of the current line

CODE:

```
E: > ass1.cpp > main()
#include <iostream>
using namespace std;

int main() {
    cout << "This is line 1\n";
    cout << "This is line 2" << endl;
    cout << "Column1\tColumn2\tColumn3\n";
    cout << "This will make a beep sound\a\n";
    cout << "1234567890\rHELLO\n";
    return 0;

}
```

OUTPUT:

```
PS E:\> cd "e:\" ; if ($?) { g++ ass1.cpp -o ass1 } ; if ($?) { .\ass1 }
This is line 1
This is line 2
Column1 Column2 Column3
This will make a beep sound
HELLO67890
```

(Q5) Implement namespace in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

CODE:

```
E: > ass1.cpp > main()
#include <iostream>
using namespace std;

namespace alpha {
    int num = 15;
}

namespace beta {
    int num = 45;
}

int main() {
    cout << "alpha namespace number: " << alpha::num << endl;
    cout << "beta namespace number : " << beta::num << endl;
    return 0;
}
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\HP> cd "e:\" ; if ($?) { g++ ass1.cpp -o ass1 } ; if ($?) { .\ass1 }
alpha namespace number: 15
beta namespace number : 45
PS E:\>
```

(Q6) Write a program to define a structure Student that contains 3 variables and 2 functions:

- Variables:

(a) name

(b) roll

(c) marks

- Functions:

(1) setStudentData()

(2) getStudentData()

Use loops to input and output data for 3 student object entries.

CODE:

E: > ass1.cpp > Student > getStudentData()

```cpp
#include <iostream>
using namespace std;

struct Student {
    string name;
    int roll;
    int marks;
    void setStudentData() {
        cout << "Enter student's name: ";
        cin >> name;
        cout << "Enter roll number: ";
        cin >> roll;
        cout << "Enter marks obtained: ";
        cin >> marks;
    }
    void getStudentData() {
        cout << "Name   : " << name << endl;
        cout << "Roll   : " << roll << endl;
        cout << "Marks  : " << marks << endl;
    }
};
int main() {
    Student s[3];
    cout << "Enter Details for 3 Students \n";
    for (int i = 0; i < 3; i++) {
        cout << "\nStudent " << i + 1<< "\n";
        s[i].setStudentData();
    }
    cout << "Displaying Student Records \n";
    for (int i = 0; i < 3; i++) {
        cout << "\nRecord of Student " << i + 1<<"\n";
        s[i].getStudentData();
    }
    return 0;
}
```

OUTPUT:

```
PS E:\> cd "e:\" ; if ($?) { g++ ass1.cpp -o ass1 } ; if ($?) { .\ass1 }
Enter Details for 3 Students

Student 1
Enter student's name: ELLE
Enter roll number: 56
Enter marks obtained: 89

Student 2
Enter student's name: EVA
Enter roll number: 89
Enter marks obtained: 90

Student 3
Enter student's name: ISHA
Enter roll number: 23
Enter marks obtained: 55
Displaying Student Records

Record of Student 1
Name   : ELLE
Roll   : 56
Marks  : 89

Record of Student 2
Name   : EVA
Roll   : 89
Marks  : 90

Record of Student 3
Name   : ISHA
Roll   : 23
Marks  : 55
PS E:\>
```

(Q7)

CODE:

E: > ass1.cpp > main()

```cpp
#include <iostream>
using namespace std;

struct Student {
    private:
        string name;
        int roll;
        int marks;
    public:
        void setStudentData() {
            cout << "Enter student's name: ";
            cin >> name;
            cout << "Enter roll number: ";
            cin >> roll;
            cout << "Enter marks obtained: ";
            cin >> marks;
        }
        void getStudentData() {
            cout << "Name   : " << name << endl;
            cout << "Roll   : " << roll << endl;
            cout << "Marks  : " << marks << endl;
        }
};
int main() {
    Student s[3];
    cout << "Enter Details for 3 Students \n";
    for (int i = 0; i < 3; i++) {
        cout << "\nStudent " << i + 1<< "\n";
        s[i].setStudentData();
    }
    cout << "Displaying Student Records \n";
    for (int i = 0; i < 3; i++) {
        cout << "\nRecord of Student " << i + 1<<"\n";
        s[i].getStudentData();
    }
    return 0;
}
```

OUTPUT:

```
PS E:\> cd "e:\" ; if ($?) { g++ ass1.cpp -o ass1 } ; if ($?) { .\ass1 }
Enter Details for 3 Students

Student 1
Enter student's name: ELLE
Enter roll number: 56
Enter marks obtained: 89

Student 2
Enter student's name: EVA
Enter roll number: 89
Enter marks obtained: 90

Student 3
Enter student's name: ISHA
Enter roll number: 23
Enter marks obtained: 55
Displaying Student Records

Record of Student 1
Name   : ELLE
Roll   : 56
Marks  : 89

Record of Student 2
Name   : EVA
Roll   : 89
Marks  : 90

Record of Student 3
Name   : ISHA
Roll   : 23
Marks  : 55
PS E:\>
```

(Q8) Convert program #4 from struct to class.

CODE:

```
E: > ass1.cpp > ...
#include <iostream>
using namespace std;

class test {
public:
    void cases() {
        cout << "This is line 1\n";
        cout << "This is line 2" << endl;
        cout << "Column1\tColumn2\tColumn3\n";
        cout << "This will make a beep sound\a\n";
        cout << "1234567890\rHELLO\n";
    }
};

int main() {
    test t;
    t.cases();

    return 0;
}
```

OUTPUT:

```
PS E:\> cd "e:\" ; if ($?) { g++ ass1.cpp -o ass1 } ; if ($?) { .\ass1 }
This is line 1
This is line 2
Column1 Column2 Column3
This will make a beep sound
HELLO67890
PS E:\>
```

(Q9) Discuss the differences of structures in C and structures in C++

In the C language, structures are meant purely for storing related data values — they cannot include functions.

However, in C++, a structure is more powerful: it can have both data members and member functions.

In fact, C++ treats a structure almost the same as a class, meaning we can also specify access levels like public and private inside it.

(Q10) WAP to read an integer value from the keyboard and display on screen "WELL DONE" that many times.

CODE:

```
E: > ass1.cpp > ...
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 0; i < n; i++) {
        cout << "WELL DONE" << endl;
    }
    return 0;
}
```

OUTPUT:

```
PS E:\> cd "e:\" ; if ($?) { g++ ass1.cpp -o ass1 } ; if ($?) { .\ass1 }
Enter a number: 6
WELL DONE
WELL DONE
WELL DONE
WELL DONE
WELL DONE
WELL DONE
PS E:\>
```